

$$8 \quad E_T = \frac{1}{2} m \omega^2 x_0^2$$

When particle is mid-way between the equilibrium position and an amplitude position

$$E_P = \frac{1}{2} m \omega^2 \left(\frac{x_0}{2} \right)^2 = \frac{1}{4} \left(\frac{1}{2} m \omega^2 x_0^2 \right) = \frac{1}{4} E_T$$

$$E_P : E_T = 1 : 4$$

$$E_K = \frac{1}{2} m \omega^2 \left(x_0^2 - \left(\frac{x_0}{2} \right)^2 \right) = \frac{3}{4} \left(\frac{1}{2} m \omega^2 x_0^2 \right) = \frac{3}{4} E_T$$

$$E_K : E_T = 3 : 4$$

Answer: D

9 The number of carbon dioxide molecules in one mole of carbon dioxide is the Avogadro